Namit Bhutani

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Research Interests

Computer graphics: real-time and physically based rendering, Gaussian splatting, spectral rendering, GPU performance.

Education

BITS Pilani, Hyderabad Campus Aug 2022 – May 2027 (expected)
B.E. (Hons.) Computer Science and M.Sc. (Hons.) Economics, dual degree *CGPA 8.92 / 10*
Relevant coursework: Computer Graphics, Machine Learning, Probability and Statistics, Algorithms, Operating Systems.

Publications

[C1] **Namit Bhutani**, Tathagata Ray, Sumanta Pattanaik. **Real-Time Spectral Photoelasticity via Stochastic Hero-Wavelength Ray Marching and Stress-Aware Temporal Reprojection.** *Pacific Graphics 2026, Conference Papers* (Eurographics), Singapore. **Oral talk and poster.** [paper] [project page] [code]

[J1] Chaitra C R, **Namit Bhutani**, Sankalp Kulkarni, Sai Rama Akash Varma Sagi, Shashank Pandey, Rohit Yalavarthy, Dipanjan Chakraborty, Prajna Upadhyay. **Generative Discourse Parsing for Indian Law.** *Artificial Intelligence and Law* (Springer), 2026. [paper]

Research Experience

Dynamic Graphics Project (DGP) Lab, University of Toronto Jun 2026 – Present
Research Intern *Toronto, Canada*

- Building a Houdini plugin and standalone application for interactive **Gaussian splatting**: brush-based splat painting over **100K+** splats at **60 FPS** (RTX 3070), with C++ ray picking over 3D Gaussians through a custom BVH and cached depth-buffer sampling.
- Distilled multi-view **DINOv3** features into per-splat embeddings (coverage-weighted fusion, frozen PCA 384→128); unsupervised segmentation answers similarity queries in under **50 ms** over 100K+ primitives.

Spectral Photoelasticity Rendering, BITS Pilani and UCF Aug 2025 – Present
Undergraduate Researcher; advisors: Prof. Tathagata Ray and Prof. Sumanta Pattanaik

- Built a physically based photoelasticity renderer in NVIDIA Falcor (C++, Slang) at **120+ FPS** (1080p) [C1]: adapted **hero-wavelength sampling** to the Jones-calculus volume integral, marching four stratified wavelengths in one packed pass, **8.55**× faster than the 16-wavelength baseline.
- Identified a temporal anti-aliasing failure for volumes that are geometrically static but optically dynamic, and designed a **stress-aware history gate** that removes ghosting under moving loads.
- Validated against a real polariscope photograph and closed-form solutions (0.4% fringe-order error); open-sourced code, scenes and scripts.

LegalTree: Discourse Parsing for Indian Law, BITS Pilani Sep 2024 – Dec 2025
Undergraduate Researcher

- Fine-tuned T5 with LoRA to parse legal sentences into typed relation trees (11 relation types), compared against few-shot GPT-4 and zero-shot chain-of-thought GPT-4o; best configuration reached **85%** accuracy. Built the evaluation harness and a RAG pipeline over 10,000+ documents [J1].

Talks and Posters

Pacific Graphics 2026, Singapore Oct 2026
Paper talk (main auditorium) and poster: Real-Time Spectral Photoelasticity

Open Source and Industry

Google Summer of Code, Blender Foundation Jun 2025 – Sep 2025
Contributor, sculpting performance *Remote*

- Sped up sculpting on **1M+ vertex** meshes with spatial cache reordering and precomputed BVHs (**+35%** throughput, **80%** faster sculpt-mode entry); built an incremental undo engine with zstd compression (**−60%** memory). Shipped in Blender after review with 10+ core developers.

Vidcentum Technologies May 2024 – Jul 2024
Software Engineering Intern *Hyderabad, India*

- Built an asynchronous search service over 50K+ documents sustaining 1,000+ requests/s at sub-100 ms latency.

Skills and Other

Graphics and GPU: C++, GLSL, Slang, NVIDIA Falcor, OpenGL compute, Houdini, Blender internals, Unity
Machine learning: Python, PyTorch, DINOv3 features, LoRA fine-tuning **Tools:** Git, Linux, Nix, Docker
Project: procLand, GPU Marching Cubes terrain in OpenGL compute shaders, 100K+ polygons at 60+ FPS [code]
Leadership: Senior Developer, Student Welfare Division, BITS Hyderabad; led a team of 4 developers